Radiology

Large-Scale Validation of the Feasibility of GPT-4 as a Proofreading Tool for Head CT Reports

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Conflicts of interest are listed at the end of this article.

See also the editorial by Choi in this issue.

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Background: The increasing workload of radiologists can lead to burnout and errors in radiology reports. Large language models, such as OpenAI's GPT-4, hold promise as error revision tools for radiology.

Purpose: To test the feasibility of GPT-4 use by determining its error detection, reasoning, and revision performance on head CT reports with varying error types and to validate its clinical utility by comparison with human readers.

Materials and Methods: A total of 10 300 head CT reports were retrospectively extracted from the Medical Information Mart for Intensive Care III public dataset. In experiment 1, among the 300 unaltered reports and 300 versions with applied errors, GPT-4 optimization was initially conducted with 200 reports. The remaining 400 were used for evaluation of error type detection, reasoning, and revision, as well as the analysis of reports with undetected errors. The performance was also compared with that of human readers. In experiment 2, the detection performance of GPT-4 was validated on 10 000 unaltered reports that were deemed error-free by physicians, and an analysis of false-positive results was conducted. A permutation test was conducted to assess differences in performance.

Results: GPT-4 demonstrated commendable performance in error detection (sensitivity, 84% for interpretive error and 89% for factual error), reasoning, and revision. Compared with GPT-4, human readers had worse factual error detection sensitivity (0.33–0.69 vs 0.89; $P = .008$ for radiologist 4, $P < .001$ for others) and took longer to review (82–121 seconds vs 16 seconds, $P < .001$). In 10 000 reports, GPT-4 detected 96 errors, with a low positive predictive value of 0.05, yet 14% of the false-positive responses were potentially beneficial.

Conclusion: GPT-4 effectively detects, reasons, and revises errors in radiology reports. While it shows excellent performance in identifying factual errors, its ability to prioritize clinically significant findings is limited. Recognizing its strengths and limitations, GPT-4 could serve as a feasible tool.

Supplemental material is available for this article.

The increasing demand for imaging has led to a greater workload for radiologists, resulting in burnout and an increase in errors on reports (1,2). The consequences of these errors, which can potentially mislead physicians and negatively impact patient care, have prompted efforts to systematically classify and reduce report errors (3). Consequently, researchers have proposed various classification schemes for radiology report errors (4–7).

The process of generating radiology reports, despite some variation, can be divided into the following two main steps: *(a)* detecting abnormalities from images and *(b)* documenting the detected abnormalities. All errors in radiology reports occur during one of these two steps. Errors in the first stage can be attributed to factors such as visual misperception or a cognitive limitation, whereas errors in the second stage may result from misinterpretation of findings or the inclusion of factually inconsistent content in the report text.

Efforts have been made to use deep learning–based natural language processing models to correct errors occurring in the second step (8–10). However, these models typically require large amounts of training data and are often designed to only handle a single report or error type, limiting their ability to be generalized across various contexts. Furthermore, these models face challenges in terms of error reasoning and revision, in addition to error detection.

Large language models (LLMs), such as OpenAI's GPT-4 (11), have simplified previously challenging tasks. In radiology, GPT-4 has shown potential in generating impressions, data mining, formatting, labeling, and speech recognition error detection (12–16), suggesting that GPT-4 may also have potential in proofreading radiology reports. However, as concerns about the reliability of these generative artificial intelligence systems persist, it is crucial to verify the accuracy of the generated response, the consistency of the model's output, and the effectiveness of its reasoning capability (17–20).

Abbreviation

LLM = large language model

Summary

OpenAI's GPT-4 could detect, reason, and revise errors in head CT reports, demonstrating its feasibility as a tool for proofreading radiology reports.

Key Results

- In this retrospective study of 10 300 head CT reports from Medical Information Mart for Intensive Care III database, OpenAI's GPT-4 was tested in two proofreading experiments.
- GPT-4 showed better factual error detection sensitivity (0.89 vs 0.33–0.69; radiologist 4, P = .008; others, P < .001) and faster review time (16 seconds vs 82–121 seconds, P < .001) than human readers.
- In 10 000 reports, GPT-4 detected 96 errors, with a low positive predictive value of 0.05; however, 14% of changes following false-positive responses were potentially beneficial.

Therefore, the aim of this study was to test the feasibility of GPT-4 use by determining its error detection, reasoning, and revision performance on head CT reports with varying error types, and to validate its clinical utility by comparison with human readers.

Materials and Methods

This retrospective study was approved by the local institutional review board. All participating human readers provided written informed consent.

Head CT Report Datasets and Study Design

This study used data extracted from the Medical Information Mart for Intensive Care III (hereafter, MIMIC-III), which is a publicly accessible database comprising reports from 53 150 patients admitted to the intensive care unit at Beth Israel Deaconess Medical Center (21). It is one of the most extensively validated open-source databases, with numerous version updates ensuring quality control, making it an optimal dataset for evaluating real-world settings in radiology report research. Details of the inclusion and exclusion criteria are illustrated in Figure 1. For this study, head CT reports were selected, excluding descriptions related to procedures. The database has a column labeled "ISERROR," allowing the identification of notes classified as erroneous by physicians (22). On the basis of these data, only reports that were verified to be free of errors were included in the study.

The design of the two study experiments is shown in Figure 2, using datasets with distinct characteristics. Experiment 1 aimed to optimize the model and analyze its performance using an error-applied dataset. The use of this error-applied dataset offers several advantages over the use of a real-world dataset, which would typically have a markedly lower error rate. This error-applied dataset was constructed to enable a comprehensive exploratory analysis of different error types and reports with undetected errors, thus identifying the model's strengths and weaknesses. Specifically, diverse errors were applied to 300 randomly selected head CT reports. As a result, a total of 600 reports were obtained (300 unaltered reports and 300 error-applied reports). In this study, the types of errors that occurred when writing the report were categorized into interpretive and factual errors (9). Error definitions and applications are detailed in Table 1 and Appendix S1. Experiment 2 aimed to validate the utility of the model in real-world reporting by evaluating its performance on 10 000 randomly selected original head CT reports.

Experiment 1 for GPT-4 Optimization

For experiment 1, 200 reports out of the full set of 600 were used for an initial optimization stage; specifically, a set of 100 unaltered reports and 100 error-applied reports were selected. Two prompts with different levels of autonomy (prompt 1 [higher autonomy] and prompt 2 [lower autonomy]) were compared. Additionally, the optimal "temperature" parameter setting, which governs the diversity of the model's output (23), was determined. The prompt design and experimental details are provided in Appendix S1 and Table S1.

Experiment 1 as Exploratory Analysis of GPT-4 Performance

*GPT-4 proofreading performance evaluation.—*The performance evaluation of GPT-4 used the remaining 400 reports (200 unaltered and 200 error-applied) to compare error detection across different error types. For report details, see Tables S2 and S3. Detected errors were assessed on the basis of their reasoning and revision quality using a five-point Likert scale. Comparative analysis was used to examine the reports with detected and undetected errors. Additionally, the inference time of GPT-4 per session was recorded.

*Comparative study with human readers.—*Human readers were tested with one of two sets of reports. These sets were derived from the same set of 400 reports evaluated by GPT-4. Specifically, either

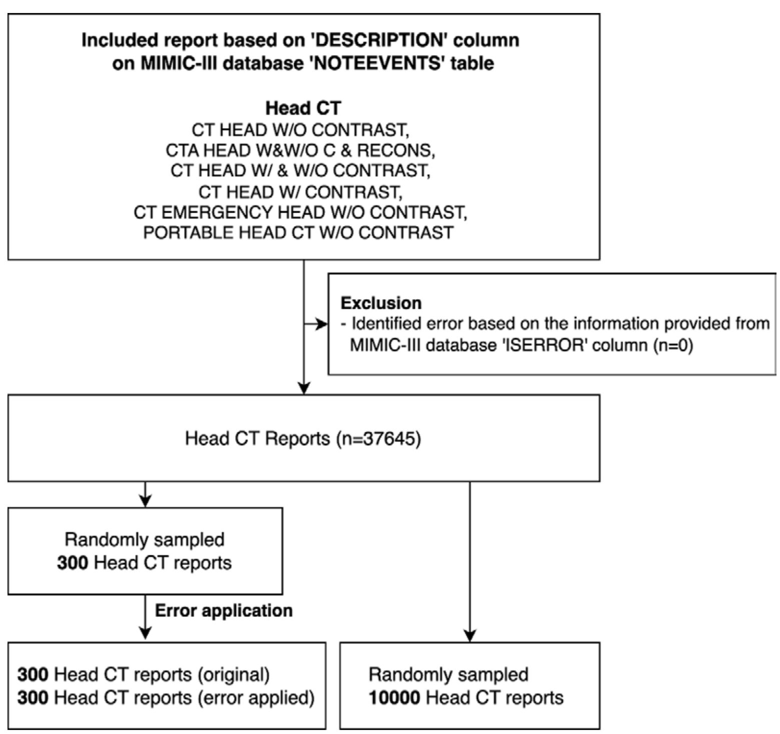


Figure 1: Flowchart shows inclusion and exclusion criteria for head CT reports extracted from the Medical Information Mart for Intensive Care III (MIMIC-III) database. C = contrast, RECONS = reconstruction, W = with, W/O = without.

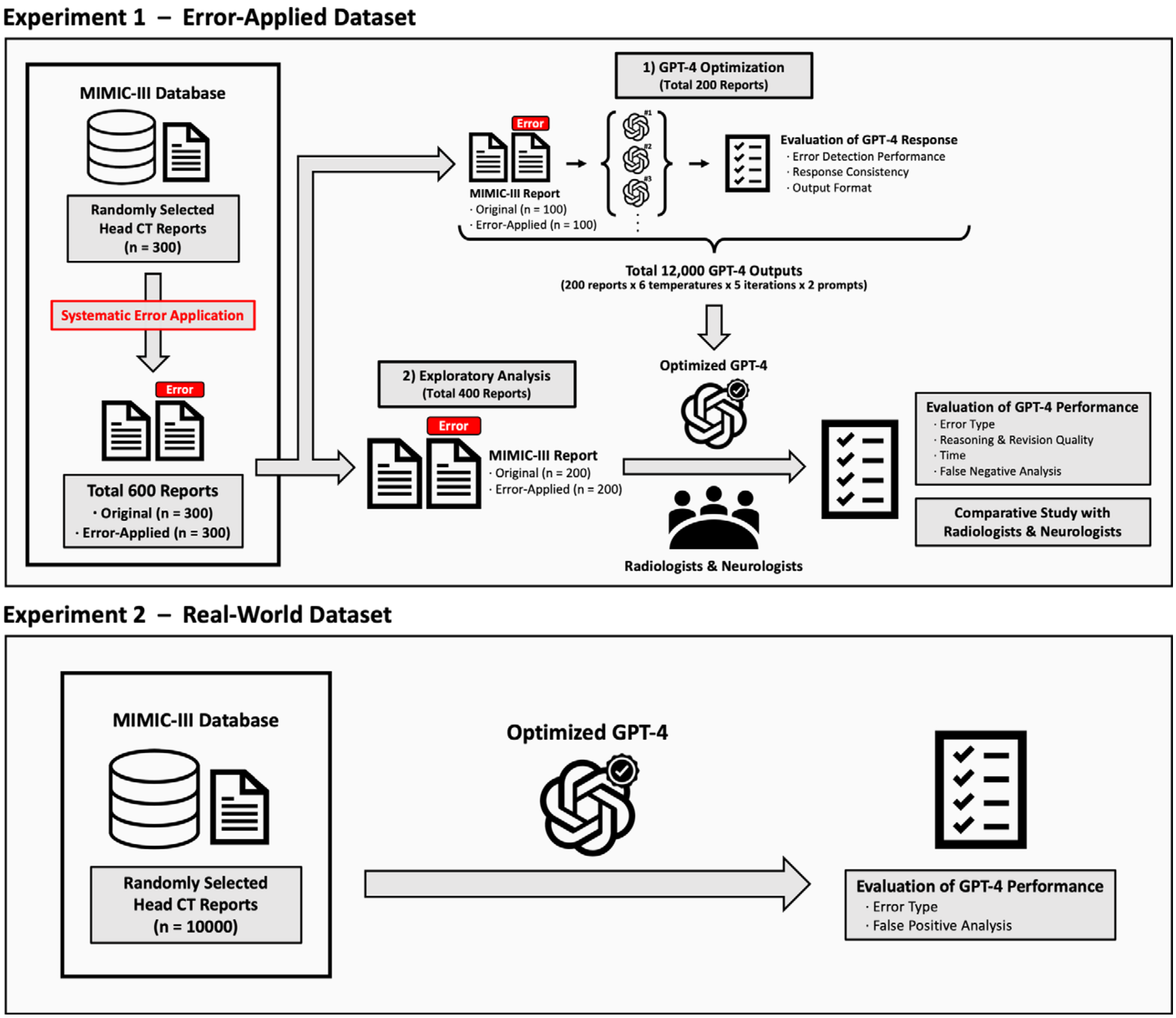


Figure 2: Schematics show the overall structure of the study using OpenAI's GPT-4, including experiment 1 (top) and experiment 2 (bottom). MIMIC-III = Medical Information Mart for Intensive Care III.

the unaltered or the error-applied version of each report was randomly selected to be included in set A or set B, resulting in two sets of 200 reports each (Fig S1). Because the original and error-applied reports differed only in the section where the error was introduced, this setup aimed to reduce recall bias. Details of the report set selection are provided in the "Report set selection for human readers" section of Appendix S1 and Table S4. Eight readers who were blinded to any information beyond the instructions proofread either set A or set B of the reports; the readers included three resident radiologists (radiologists 1, 2, and 3) with 1, 2, and 4 years of experience, respectively; two neuroradiologists (radiologists 4 and 5) with 5 and 15 years of experience, including 1 and 11 years in neuroradiology, respectively; and three neurologists (neurologists 1, 2, and 3) with 5, 6, and 15 years of experience, respectively. A 3-minute time limit was set for reviewing each report (Fig S2).

Experiment 2 for Validation on a Real-World Dataset

Experiment 2 used 10 000 unaltered reports. Reports flagged as having errors by GPT-4 were reviewed and sorted as either true errors or false-positive results. The clinical impact of false-positive responses was further assessed to determine their actual harmfulness or potential benefits, with the detailed criteria outlined in Table S5. The results of the review and evaluation tasks were evenly distributed between the two radiologists.

Ad Hoc Analysis of Few-Shot Prompts for Decreasing the False-Positive Rate

An ad hoc analysis was conducted to investigate the potential ability of few-shot prompts to reduce the false-positive rate. The detailed methodology is available in Appendix S1.

Statistical Analysis

Data normality was confirmed with the Shapiro-Wilk test. Sensitivity, specificity, F1 score, and accuracy were used to evaluate the error detection performance. The Mann-Whitney *U* test was used to compare the word, sentence, and impression counts of the reports with detected and undetected errors. Performance differences between GPT-4 and human readers were evaluated

Table 1: Types of Errors, Their Descriptions, and Examples Used in This Study

Error Type	Description	Example: Finding	Example: Impression
Interpretive error			
Substitution (1a)	Incorrect interpretation or misclassification in the Impression section about a specific finding mentioned in the Finding section.	There is no intracranial hemorrhage, edema, mass effect or vascular territorial infarction. Ventricles and sulci are enlarged, and normal in configuration consistent with global parenchymal volume loss.	*No acute cardiopulmonary abnormality.*
Omission (1b)	Underreporting or missing the presence or severity of a specific finding in the Impression section compared with the Finding section.	There is no evidence of hemorrhage, edema, mass effect, or infarction. The ventricles and sulci are normal in size and configuration. *There is a small left parietal subgaleal hematoma (series 3, image 46).*	*Normal study.*
Addition (1c)	Overreporting the presence or severity of a specific finding in the Impression section compared with the Finding section.	*There is no evidence of mass lesion or abnormal enhancement.* There is no shift of the normally midline structures, hydrocephalus, or evidence of infarction. An ET tube and nasoenteric tube are noted. A moderate amount of secretions are seen within the posterior nasopharynx.	*A rounded small enhancing lesion noted in the right posterior communicating artery distribution.*
Factual error			
Discrepancy in location (2a)	Discrepancy in location of the lesion between the Finding and Impression sections	The patient is *status post left frontotemporal craniotomy* with normal postoperative changes of mild pneumocephalus and fluid at the operative site. No definite intracranial hemorrhage is seen. Along the right temporal and frontal aspect of the skull, there is a *large extracranial soft tissue density likely representing hematoma and swelling.*	Status post *right* frontotemporal craniotomy with evacuation of subdural hematoma. No evidence of intracranial hemorrhage. Large right *extra-axial* hematoma.
Discrepancy in the numerical measurement (2b)	Discrepancy in numerical measurement between the Finding and Impression sections	Compared with prior study, there has been decrease in the size of previously seen right-sided subdural hematoma, now measuring only approximately *2 cm* compared with its prior measurement of 7 mm. There is a *4.6 × 3.7 × 5 cm* (transverse, AP and craniocaudal dimensions, respectively) mass lesion with a broad base to the adjacent left frontal skull, could be consistent with the given history of meningioma.	Interval decrease in size of both right and left subdural hemorrhages, now measuring approximately *2 mm* in maximal depth on the right side. *4.1 × 3.8 × 5 cm* left frontal mass lesion arising from the left frontal skull/dura consistent with the given history of meningioma.

Note.—The examples presented in the table are actual errors from the original Medical Information Mart for Intensive Care III head CT reports detected by GPT-4 in this study, with the error portions in italics. AP = anteroposterior, ET = endotracheal.

across sensitivity, specificity, positive predictive value, negative predictive value, accuracy, F1 score, and time spent for a report. A permutation test with 1000 permutations was conducted to calculate the 95% CI (24); two-sided *P* values were then derived by comparing the observed difference with the null distribution quantiles. The threshold for significance was set at a *P* value of .05. Statistical analyses and preparation of graphics were performed in Python (version 3.11.4; Python Software Foundation) using the Pandas (version 2.1.1) and Matplotlib (version 3.4.2) packages.

Results

Experiment 1 and GPT-4 Optimization

Compared with prompt 1, prompt 2 showed higher sensitivity (range, 0.85–0.89 vs 0.77–0.81; $P < .001$) but lower specificity (range, 0.71–0.81 vs 0.85–0.93; $P < .001$) (Fig S3A). The F1 score peaked when the temperature was set to 0.2 (Fig S3B). The results are detailed in Tables S6 and S7.

Experiment 1 and Exploratory Analysis of GPT-4 Performance

Error detection.—The error detection performance of GPT-4 was evaluated on 400 reports. The detection sensitivity was 0.86, the specificity was 0.76, and the F1 score was 0.82. The detection sensitivity was 89% for factual errors and 84% for interpretive errors (Tables 2, S8). The detection sensitivity was lower for substitution errors (type 1a, 76%) and omission errors (type 1b, 78%) (Fig 3A).

Error reasoning and revision.—GPT-4 showed a strong ability to reason detected errors, achieving a Likert rating of 4.96 out of 5 with interpretive errors and 5 with factual errors. It also displayed satisfactory revision ability, scoring 4.45 out of 5 with interpretive errors and 5 with factual errors (Fig 4).

Table 2: Detailed Metrics of Proofreading Performance between GPT-4 and Human Readers

Metric	GPT-4	Radiologist 1	Radiologist 2	Radiologist 3	Radiologist 4	Radiologist 5	Neurologist 1	Neurologist 2	Neurologist 3
Sensitivity									
Total	0.860	0.618*	0.676*	0.731†	0.784	0.787	0.676*	0.596*	0.764
Type 1 overall	0.838	0.771	0.884	0.854	0.841	0.979†	0.768	0.771	0.897
Type 2 overall	0.892	0.439*	0.333*	0.585*	0.690‡	0.561*	0.524*	0.390*	0.516*
1a	0.757	0.786	0.957	0.786	0.783	0.929	0.870	0.714	0.857
1b	0.775	0.556	0.682	0.833	0.773	1†	0.455†	0.667	0.778
1c	0.975	1.000	1.000	0.938	0.958	1	0.958	0.938	1
2a	0.904	0.480*	0.481*	0.600‡	0.741	0.52*	0.519*	0.360*	0.52*
2b	0.871	0.375‡	0.067*	0.563†	0.600	0.625	0.533†	0.438‡	0.5
Specificity	0.755	0.771	0.787	0.734	0.967*	0.901‡	0.843	0.838	0.856†
PPV	0.778	0.688	0.798	0.691	0.967*	0.864	0.676	0.746	0.81
NPV	0.844	0.712‡	0.660*	0.770	0.782	0.84	0.676‡	0.721†	0.819
Accuracy	0.808	0.702‡	0.725†	0.732†	0.865	0.85	0.750	0.730†	0.815
F1 score	0.817	0.651*	0.732†	0.710‡	0.866	0.824	0.750	0.663‡	0.786
Time per report (sec)	15.5	81.9*	100.6*	93.3*	95.9*	97.8*	94.0*	120.6*	94.5*

Note.—A permutation test with 1000 permutations was conducted to calculate the 95% CI for comparing OpenAI's GPT-4 and the human reader across each metric, with two-sided *P* values derived by comparing the observed difference to the null distribution quantiles. NPV = negative predictive value, PPV = positive predictive value.

* Indicates a statistically significant difference from GPT-4, $P < .001$.

† Indicates $P < .05$.

‡ Indicates $P < .01$.

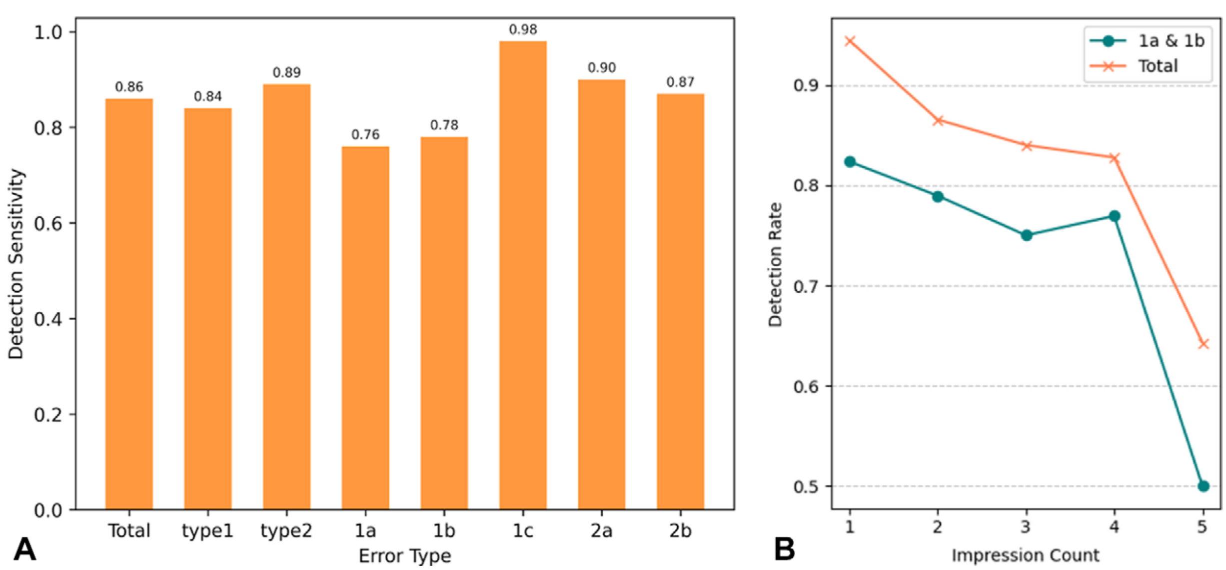


Figure 3: **(A)** Bar graph shows a comparison of GPT-4 (OpenAI) error detection sensitivity for each error type. **(B)** Line graph shows GPT-4 detection sensitivity for various impression counts.

Comparative analysis of the reports with detected and undetected errors.—The word and sentence counts were higher for reports with undetected errors, but not significantly (word count: 299 vs 338, $P = .051$; sentence count: 23 vs 26, $P = .13$). The impression count was significantly greater for reports with undetected error (2 vs 3, $P = .01$) (Fig S4). As the impression count increased, the detection sensitivity decreased, particularly for type 1a and 1b interpretive errors. The detection sensitivity decreased from 82% to 50% as the impression count increased from 1 to 5 (Fig 3B, Table S9).

Measurement time.—GPT-4 took 15.5 seconds per task on average, totaling 6210 seconds for 400 reports.

Comparative study with human readers.—The detection sensitivity of radiologist 5 for interpretive errors was significantly greater than that of GPT-4 (0.98 vs 0.84, $P = .02$). There was no evidence of a difference in interpretive error detection sensitivity between the other human readers and GPT-4. All human readers had lower factual error detection sensitivity, showing statistically significant differences to GPT-4 (0.89 vs 0.33–0.69; $P = .008$ for radiologist

4, $P < .001$ for others). Neuroradiologists (radiologists 4 and 5) demonstrated significantly greater specificity than did GPT-4 (0.90–0.97 vs 0.76; $P < .001$ for radiologist 4, $P = .004$ for radiologist 5). The F1 score of radiologist 4 was greater than that of GPT-4, but this difference was not statistically significant (0.87 vs 0.82, $P = .87$). The time differences between each of the human readers and GPT-4 were statistically significant (82–121 seconds vs 16 seconds, $P < .001$). Detailed comparisons are provided in Table 2 and Figure 5.

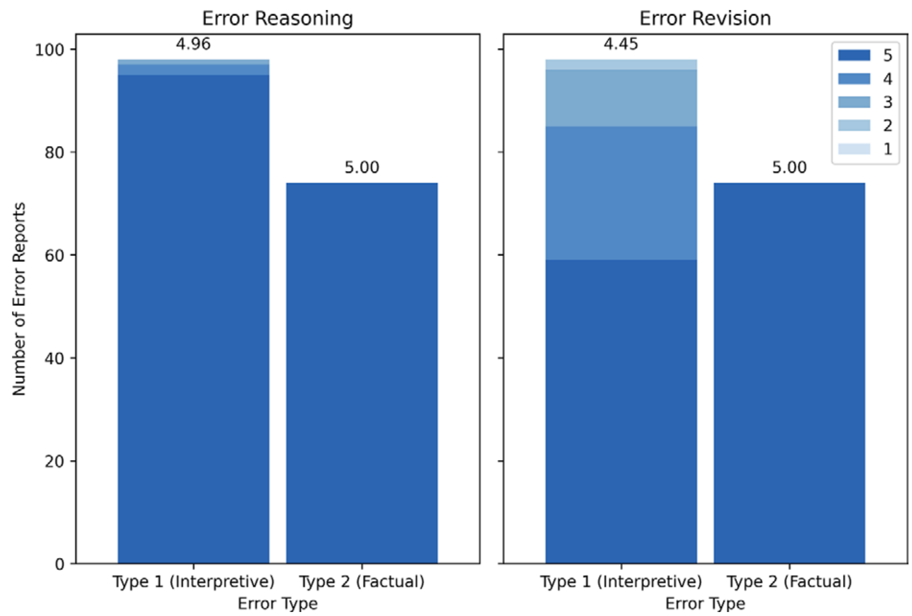


Figure 4: Bar graphs show the quality evaluation of error reasoning (left) and error revision (right) capabilities by GPT-4 (OpenAI) using a five-point Likert scale. A score of 5 indicates an excellent response.

Experiment 2 and Validation on a Real-World Dataset

Error detection.—Among the 10 000 reports analyzed, GPT-4 classified 1792 reports as having errors, among which there were 96 true errors, yielding a positive predictive value of 0.054. The types and counts of errors identified in the real-world dataset are detailed in Figure 6A and Table S10. The most prevalent error type was 2a ($n = 51$), with the most common issue being incorrect laterality ($n = 37$). Furthermore, GPT-4 successfully identified at least one error of every type. All the detected true errors are documented in Table S11.

Clinical impact of false-positive results.—The clinical impact of false-positive results incorrectly classified by GPT-4 is presented in Figure 6B and Table S12. The rate of erroneous false-positive results, at only 1% of all false-positive results, was very low. These false-positive results included reports where the "evolution" of a hemorrhage into a subacute stage or an increase in the density of a mass was misinterpreted as disease progression. There were no reports for which the model created errors on the basis of content not in the report.

In the analysis of false-positive results, 239 report changes (14%) were identified as potentially beneficial. The improvements made were as follows: correction of typographical and grammatical errors ($n = 14$); addressing oversimplified impressions sections, thereby making the reports more understandable ($n = 94$); provision of additional details for observations that were mentioned in the impressions section of the report but not adequately elaborated in the findings section, ensuring more thorough documentation ($n = 30$); and addition of clinically significant details such as hemorrhages, infarcts, tumors, or newly observed findings, which were present in the findings section of the report but omitted from the impressions section, thus making the impressions section more comprehensive ($n = 101$).

The majority of serious false-positive results involved the addition of clinically insignificant observations to the impressions section of the report. These additions did not enhance clarity or coherence, but instead unnecessarily lengthened the impressions section. Although there were many instances of serious false-positive results ($n = 1016$), the revisions were generally confined to a few categories, with typical examples including findings such as small vessel disease, nasal sinus abnormalities, or other chronic lesions that are less important in a clinical context.

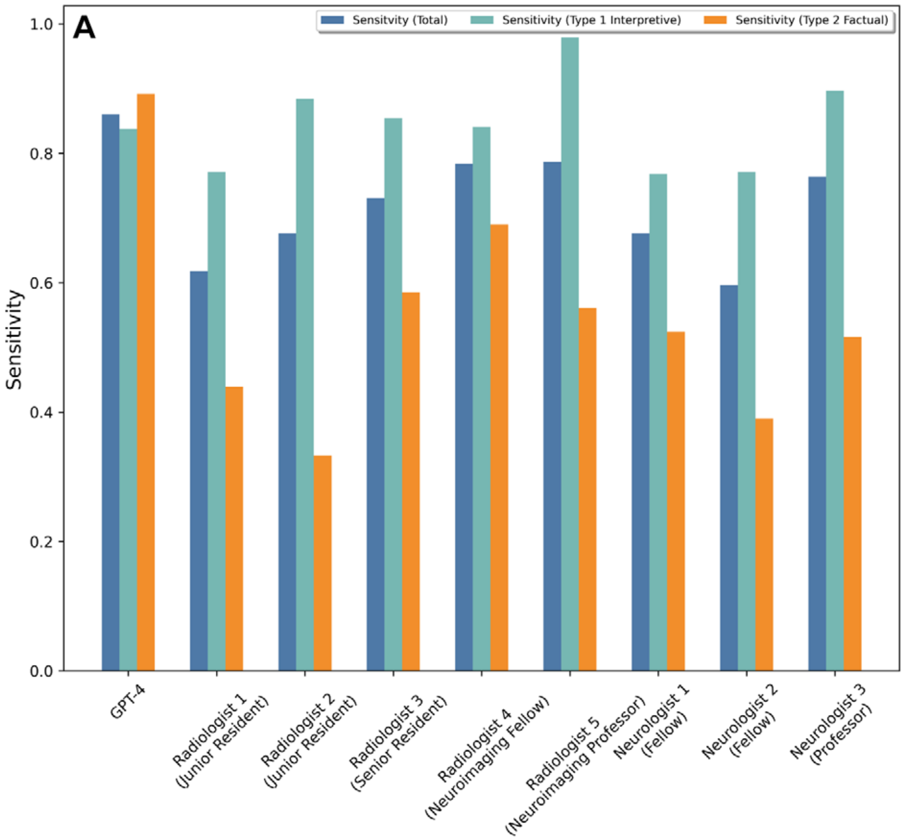
Ad Hoc Analysis and Few-Shot Prompts for Decreasing the False-Positive Rate

The positive predictive value increased from 0.06 in the original zero-shot prompt to 0.11 in the 5-shot prompt. The complete results are available in Appendix S1, Figure S5, and Table S13.

Discussion

The increasing workload of radiologists can lead to a higher rate of errors in radiology reports. In our study, OpenAI's GPT-4 (11) demonstrated commendable performance in the detection, reasoning, and revision of errors in reports of head CT. Its detection sensitivity was greater for factual errors than for interpretive errors (89% vs 84%). The sensitivity decreased as the number of impressions in the report increased. In 10 000 reports confirmed to be error-free, GPT-4 identified errors of approximately 1%. Although there was a high incidence of false-positive results (positive predictive value, 0.054), the rate of erroneous false-positive results was only 1%, and 14% of the false-positive results led to potentially beneficial report revisions. Human readers showed generally comparable performance in detecting interpretive errors, but their detection of factual errors was significantly lower and interpretation took longer.

Using GPT-4 for proofreading offers advantages over traditional deep learning models. GPT-4 showed good performance without the need for further training, thereby demonstrating its good generalizability. In the study using unaltered reports from the MIMIC-III public database that were confirmed to be error-free by radiologists and physicians, GPT-4 detected errors at a rate of 0.96% in 10 000 reports, successfully identifying a wide range of error types. Among the errors found, there were laterality errors in 0.37% of all reports; this rate was approximately eight times higher than the previously reported rate of 0.05% (25,26). Additionally, the model showed proficiency in reasoning and revising errors, demonstrating the explainability of the model, which is crucial



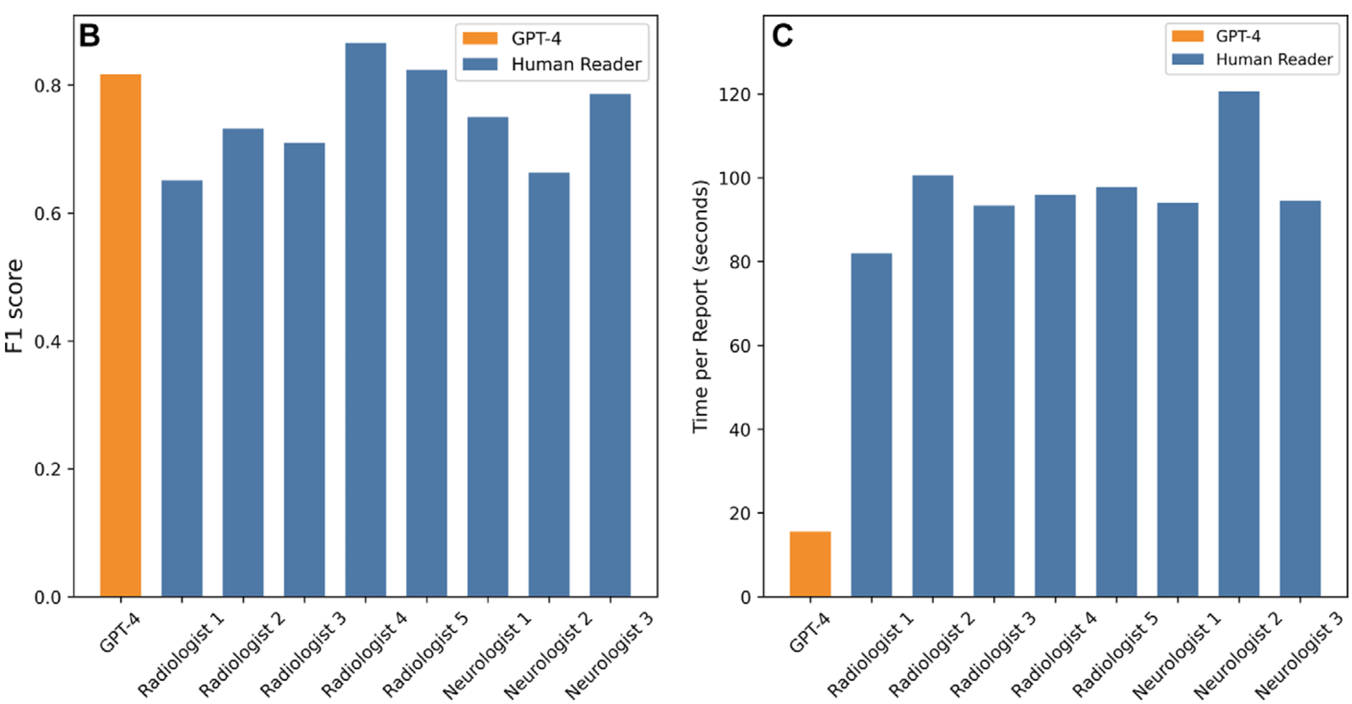


Figure 5: Bar graphs show a comparison of proofreading performance between human readers and GPT-4 (OpenAI) for **(A)** detection sensitivity across different error types, **(B)** F1 scores, and **(C)** the time spent proofreading each report.

for workflow enhancement. The study also evaluated the risk of harm of implementing the model and reported a low rate of erroneous false-positive results, indicating a low risk of harm. Finally, GPT-4 stands out for its speed and cost-effectiveness, processing a MIMIC-III report containing an average of 300 words in approximately 15.5 seconds at a cost of less than $0.01 per report (27,28).

This study also identified a major weakness of GPT-4; that is, its struggle to prioritize clinically significant findings among multiple observations. The impression section of a radiology report is not merely a summary but also an assessment of the order of importance of the findings made by the radiologist. While GPT-4 excelled at identifying factual consistency, it failed to prioritize the clinical significance of multiple findings. Consequently, with an increase in the number of impressions, GPT-4 faced challenges in coherently organizing these elements, ultimately leading to lower detection sensitivity and a higher rate of false-positive results.

Understanding a model's strengths and weaknesses is key to evaluating its clinical utility. For example, Mehta (29) highlighted the need to recognize the differences between human and computer vision to optimize cancer detection. Humans excel in adapting to and managing complex scenarios, whereas computers are better at processing speed and object recognition. This study showed that these strengths are also relevant to error detection in reports. Human readers interpret the report by understanding the nuances of the patient's clinical scenario and prioritizing the clinical importance of each finding. As a result, detecting interpretive errors may be easier for humans. However, factual errors can occur without disrupting the coherence of the surrounding sentences and often result from minor character changes. As a result, even experienced physicians may have a lower detection sensitivity for these factual errors. In contrast, GPT-4 is highly efficient at detecting factual errors, doing so significantly faster because of its ability to process large data volumes without fatigue or introducing psychologic biases.

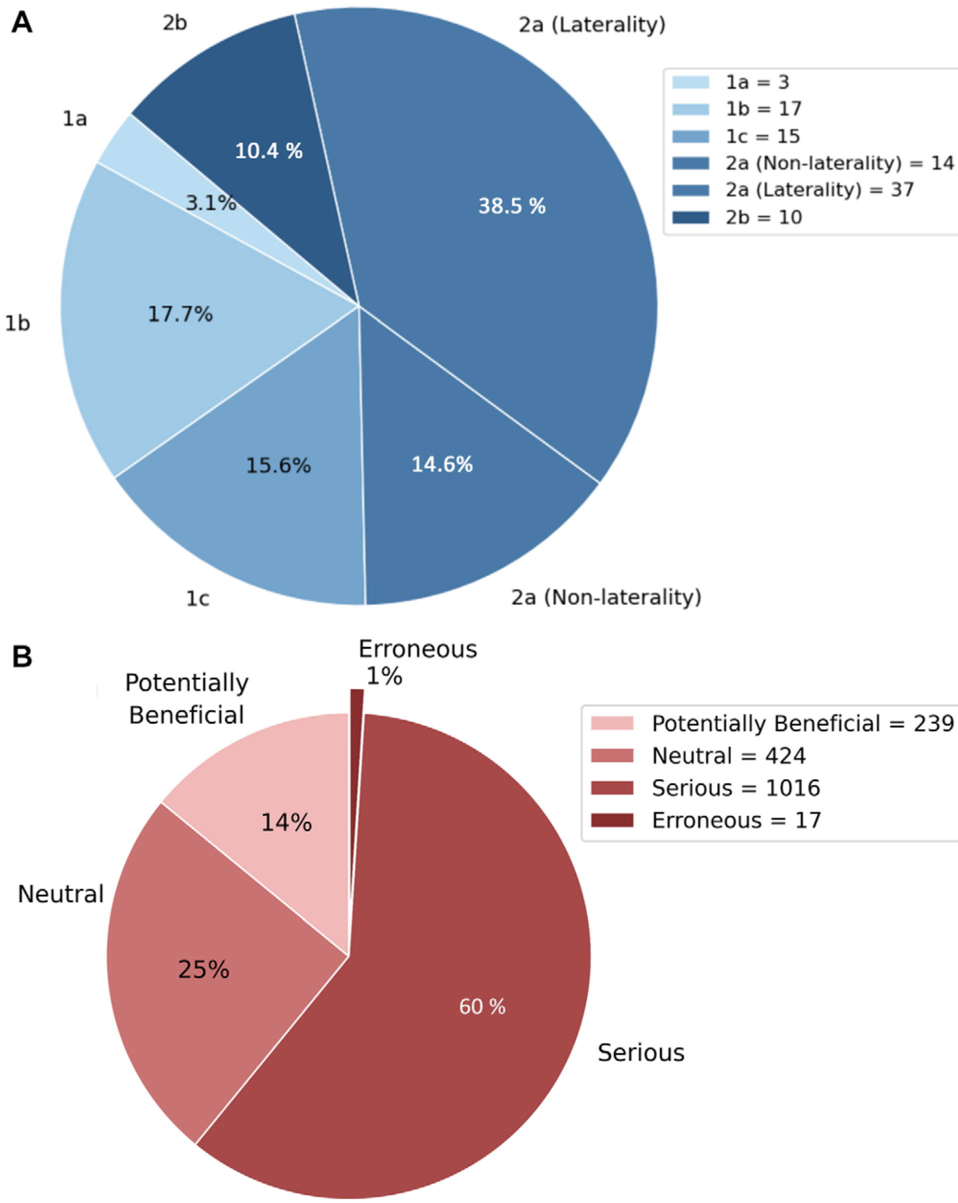


Figure 6: Reports classified as containing errors by GPT-4 (OpenAI) among 10 000 head CT reports. **(A)** Pie chart shows true errors detected by GPT-4, categorized according to error type. **(B)** Pie chart shows false-positive responses generated by GPT-4, categorized according to clinical impact.

This study highlights the possibility of synergistic collaboration between radiologists and LLMs in report proofreading. These models are not only capable of identifying factual inaccuracies but also assist in handling the tedious aspects of improving the clarity and comprehensiveness of the reports and correcting typographical and grammatical errors. With reassurance from these models, radiologists can concentrate solely on visual inspection, intellectual interpretation, and clinical judgment. Despite GPT-4's high false-alarm rate, which poses challenges to application, its implementation for proofreading resident's preliminary reports may be feasible nonetheless, given the significantly higher error rates compared with the staff-confirmed final reports (20). Intervention by GPT-4 in the early stages of proofreading could greatly lessen the workload of both the trainee and the attending radiologist.

On the basis of our research, future studies should focus on addressing the limitations of this LLM by applying additional prompt design techniques or curating a task-specific dataset for fine-tuning the model. Exploring the application of various foundation models, ranging from open-source LLMs to large multimodal models and to diverse modalities and anatomic sections would increase the generalizability of the results. Moreover, integrating the various strengths of these artificial intelligence models, such as automatic structured report generation, should be investigated as the most practical and promising aspect of collaboration between artificial intelligence and radiology, potentially streamlining the radiology workflow and improving overall efficiency. The rapid developments of LLMs point to a transformative future in radiology, with the potential to increase efficiency, reduce workload, and allow accurate reporting.

This study had several limitations. First, the exploratory analysis in experiment 1 was conducted in a simulated environment that was not identical to real-world clinical practice, and various factors may have influenced the readers' performance. Factors that could have led to an underestimation of the reader performance include the need to proofread reports written by others, a higher proportion of errors in the studied reports than in the real world, the need to proofread relatively lengthy reports, a lack of corresponding images, instructions that were optimized for GPT-4 rather than the human participants, and the 3-minute time limit. Conversely, factors that could have led to overestimating the reader performance include the inherent bias in the error-finding task and the learning effects from detecting similar errors multiple times. These factors should be carefully considered when interpreting the results of our study. Second, there may have been sampling and selection bias in compiling the MIMIC-III reports. To mitigate these limitations, the disease categories of all the reports were manually labeled to increase diversity. Third, the error taxonomy and definitions of false-positive subtypes used in this study could be considered subjective. To avoid overestimating or underestimating the performance of GPT-4, we further categorized these definitions as granularly as possible for a more accurate assessment. However, certain granular error types may still not reflect errors that occur practically, and this should be considered when interpreting the results. Fourth, we evaluated the cost-effectiveness of the model only from the perspective of using a ChatGPT service. We did not assess the costs associated with deploying the model in a local institution. Integrating such a model would require additional infrastructure, such as servers and software, to facilitate its interaction with existing systems. Future studies should consider these expenses when evaluating the cost-effectiveness of implementing the model in real-world practical scenarios.

In conclusion, OpenAI's GPT-4 effectively detected, reasoned, and revised errors in radiology reports, particularly in identifying factual errors compared with human readers. However, it struggled to prioritize clinically significant findings among multiple observations in these reports. Recognizing these strengths and limitations, GPT-4 can serve as a tool for proofreading radiology reports, thereby enhancing radiologist accuracy and efficiency in generating reports.

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